

JAD MENKARA

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EDUCATION

University of Toronto, St. George

Toronto, ON

Honours B.A.Sc, intended major in Electrical Engineering

Sept 2026 - Jun 2030 (expected)

- **Robert Little and Debra Deane Little Scholar (\$43k)** - one of 10 selected based on superior academic excellence and extracurricular involvement
- **Horatio Alger Canadian Scholar (\$10k)** - one of 150 selected based on leadership and overcoming adversity
- **Awards – National Bronze Medal** (Canada Wide Science Fair), **Sanofi Biogenius Canada Award**

TECHNICAL SKILLS

Languages: Python, Java, C/C#/C++, R

Software: Fusion 360, AutoCAD, Inventor, Solidworks, ROS2, Matlab, Simulink, Altium Designer, KiCAD, Autodesk CFD, Unity, Prusa

Slicer/generic slicers, Linux, VIM, Claude Code

Technologies: OpenCV, Pytorch, Tensorflow, Keras, Docker, Github, React, Flask, MongoDB, Bootstrap, Tailwind

Hardware: Arduino, ESP32, STM32 Architectures, Raspberry Pi, PCB Design, Sensors & Actuators, 3D Printing, Soldering

Specialties: Hardware, CAD Design, PCB/EDA Design, C++ embedded programming, simulation software, computer vision

EXPERIENCE

Robotics Engineering Intern

Remote | Calgary, AB

OrbSurgical LTD

May 2026 – Present

- Joining team designing surgical robotics systems in ROS2, Simulink, and various CAD software
- Creating a system-wide simulation of a novel surgical robot using ROS2 and Simulink on a Linux system to simulate and predict robot movements under control conditions
- Prototyped and manufactured a novel continuum robot for surgical applications (details under NDA)

Medical Robotics Engineering Research Assistant & Lead Project Developer

Remote | Calgary, AB

neuroArm Medical Robotics Laboratory @ UCalgary

Apr 2024 – May 2026

- Iterated over 30 prototype designs, designing robot components in Fusion 360, and performing static stress FEA simulations to predict deformation under load for a novel neurosurgical robot prototype aimed at removing inner-brain tumors
- Designed **schematic and board layout in KiCAD** for a **4-layer PCB** incorporating multiple voltage levels (3.3V-12V) and signal protocols (**I2C, SPI, UART, etc**); iterated over 5 times; designed for compact size constraints and ease-of-use
- Was invited to an **all-expense-paid trip to UCalgary** where I toured the neuroArm Lab and presented my work in-person to multiple neurosurgeons, engineers, and researchers

Robotics Engineering Research Assistant

Windsor, ON

IMPACT Engineering Lab @ University of Windsor

Jul 2025 – Present

- Designed, manufactured, and tested a novel, compliant, **soft-robotic gripper** with **Fusion 360** and **FDM 3D printing** for agricultural automation through gripping produce
- Performed silicone mold making and material testing on gripper components to improve compliance and range of fruit handling
- **Developed a Python program** to optimize design parameters for a wide variety of agricultural handling applications
- Currently writing manuscript for **upcoming 1st author publication submission to a peer reviewed journal** to showcase results

Avionics Design Team Member

Remote | Atlanta, GA

Propulsive Landers Rocket Design Team @ Georgia Institute of Technology

Sep 2025 – Present

- Designed **3, 4-layer PCBs with multiple flood planes** for different voltage levels (**3.3V-24V**) and signaling protocols using **Altium Designer** and **KiCAD**, minimizing temperature rise for a prototype hybrid rocket aimed at being the first in the world to take off and land repeatably

PROJECTS

Qualcomm Award (\$2400 value) @ Hack The 6ix - Autonomous Fruit Picker Robot | Arduino Uno Q, LiDAR+SLAM Jul 2026

- Designed a 3D-printed, 400cm, 4-DOF robotic arm capable of holding fruits 3-70mm in diameter, and up to 1kg, for <\$100
- Implemented seamless, low latency communication between Raspberry Pi 5 (Ubuntu), and the Arduino Uno Q (Debian), using a Flask server in under 36 hours
- Designed a rover, 100cm in length, 50cm in width, capable of carrying up to 10kg for under \$150, and run autonomously using a combination of 2D LiDAR+SLAM and Computer Vision (custom trained fruit-detection algorithm) running in parallel

MasseyHacks 2nd Place - GCI Dementia Gauntlet | Fusion 360, Python, JS, OpenCV, Flask, Py3TTSx, MySQL May 2025

May 2025

- Built a wearable gauntlet with face recognition using **Raspberry Pi**, **OpenCV**, and vibration motors for haptic feedback
- **Designed and 3D-printed a custom chassis** and integrated embedded electronics with PWM control for seamless interaction
- Developed a **Flask-based** web app for live visual feedback, incorporating a server and database

WRSTEF Gold - EMG-Based Bionic Prosthesis | MicroPython, Fusion 360, KiCAD, 3D-printing Jul 2023 – Mar 2024

Jul 2023 – Mar 2024

- Designed a modular **CAD design in Fusion 360** via joints and master sketch techniques – manufactured design via 3D-printing
- Engineered a custom tendon-driven finger actuation system, fabricated via **FDM 3D-printing**
- Designed and soldered a 20A high-current PCB with SZBK07 buck converter and **Raspberry Pi Pico MCU**
- Designed and soldered a custom PCB in KiCAD for optimal electrical performance and signal communication with peripherals